

Generative LLM and proposal writing in Scientific Communication and code generation

“[Leveraging AI tools to develop the writer rather than the writing.](#)” outlines all the different steps where generative LLM tools can help in the writing phase.

So to accommodate the different approaches to allowing generative LLM tools, and to increase transparency in the process, each student will, for every graded course component:

- discuss with their advisor(s) which of the applications outlined in Ingley and Pack (2023) they are allowed to use in the development of their proposal
- Submit this agreement as an assignment for the course
- For the applications they are allowed to use, explicitly identify how this will be transparently identified.

For instance, if a student can use ChatGPT to improve the grammar and syntax of their proposal, they will first have to write the draft themselves, then use some of the prompts in Table 1, show the suggestions from Chat GPT, and finally which ones they did and did not incorporate into their updated draft.

In addition, advisor(s) will have to explain to their students what coding tools are acceptable and/or required, and how they should be used. This is probably even more case dependent compared to how generative LLMs can be used in the writing process, and we will leave this completely open to the advisor's discretion.

Table 1 Example applications of generative AI tools

Application	Example initial prompt	Example follow-up prompt
Exploring scientific terminology	You are a science education teacher. Create a list of the most important scientific terms related to ____ and break them down into their prefixes, roots, and suffixes.	Now put these words into a table and arrange them from most common to least common.
Brainstorming	I would like to do some brainstorming for my thesis topic. The general topic I'm interested in is _____. Could you please take on the role of a research adviser and ask me questions that will help brainstorm a project?	Can you take the ideas we've brainstormed and consolidate them into bullet points?
Experimental design	I need to design an experiment. I want you to take on the role of a research adviser who will prompt me in developing my ideas and designing my experiment. I am interested in testing _____. Don't tell me how to design my experiment, but ask questions to guide my decision-making process.	What are some of the possible problems I may run into with this sort of research design?
Outline generation	You are a manuscript reviewer for a scientific conference/journal. I need to generate an outline for a (grant/manuscript/proposal). Could you please provide me with guidance on what information I should include in the outline?	Here are my ideas in an outline format. Can you offer suggestions on how to improve the organization of the paper? Explain your reasoning. How do your suggestions actually improve the paper?
Enhancing clarity and coherence	You are an editor of a scientific journal. I have a draft of a/an (abstract/manuscript/proposal, etc.). Please review this draft and suggest where the clarity and/or coherence could be improved.	Explain why you made these suggestions. Explain how your suggestions make the draft more clear/coherent.

Improving grammar and syntax	You are a professional English teacher. I have a draft of a/an (abstract/manuscript/proposal, etc.). Please review this draft and suggest where the grammar and/or syntax could be improved.	Consolidate the errors I've made in grammar and syntax into a table. In addition to identifying the errors I've made in the table, explain why what I've written is wrong and how to fix it.
Improving writing style	You are an editor for a scientific journal. I have a draft of a/an (abstract/manuscript/proposal, etc.). Please review this draft and suggest where the writing style could be changed to conform with ____ writing style.	Identify language I use that detracts from the academic quality of the manuscript and explain why these particular words or phrases detract from the academic quality.